

GNN-Enabled Trustworthy Spatio-Temporal Sensor Fusion for Smart Agriculture

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February 2026

Abstract

Smart agriculture deployments increasingly rely on dense IoT sensor networks to monitor environmental variables such as temperature, humidity, soil moisture, and light intensity. However, harsh outdoor conditions, intermittent connectivity, and low-cost hardware frequently cause sensor drift, stuck-at faults, random noise, and data loss. Conventional multi-sensor fusion and deep learning-based anomaly detection methods typically either treat sensors in isolation or neglect the spatio-temporal structure of the deployment, leading to unreliable fused measurements and cascading errors in downstream control modules such as irrigation systems.

This work proposes a GNN-enabled framework for trustworthy spatio-temporal sensor fusion in smart agriculture. We model the agricultural sensor network as a dynamic spatio-temporal graph, where nodes correspond to heterogeneous sensor streams and edges encode both physical proximity and correlation-based relationships. A spatio-temporal graph neural network (ST-GNN) jointly captures spatial topology and temporal dynamics to (i) detect and characterize sensor faults, (ii) learn node-wise credibility scores and a global system trust indicator, and (iii) drive reliability-aware fusion that suppresses low-credibility readings and imputes missing values.

To support training and evaluation, we construct multi-source datasets by augmenting open IoT benchmarks with controlled noise, drift, removal, and temporal distortion, and by collecting data from a physical smart farm testbed built from ESP32-based sensor nodes and a Raspberry Pi 4 edge gateway. The proposed framework is designed for real-time edge inference, enabling rapid response to sensor issues through adaptive weighting and sensor-control policies. We outline an experimental plan to benchmark our method against state-of-the-art deep learning and fusion baselines under diverse fault conditions, and to quantify its impact on the reliability of agricultural decision pipelines.

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1 Introduction

Smart agriculture and Agricultural Internet of Things (Ag-IoT) systems deploy dense networks of environmental sensors in fields, greenhouses, and livestock facilities to support data-driven irrigation, ventilation, and fertigation. These systems typically assume that sensor readings are trustworthy; yet in practice, Ag-IoT sensors operate in harsh, unattended environments and frequently suffer from soft and hard faults such as bias, drift, accuracy degradation, stuck-at failures, and communication outages [1]. Faulty measurements can directly trigger incorrect control actions and propagate into higher-level decision models, undermining both yield and farmers’ trust in automation.

Recent surveys on sensor fault diagnosis for Ag-IoT highlight this paradox clearly: agricultural sensors are essential for intelligent control, but their failures are common and difficult to diagnose reliably in situ [1]. For example, Liang et al. design an improved dung beetle optimization–support vector machine (IDBO–SVM) model that identifies multiple fault types (bias, drift, accuracy degradation, complete failure) in agricultural environmental sensors and achieves high classification accuracy on data collected from a Sichuan pepper farm [2]. To compensate for the scarcity of labeled faults, that work constructs mathematical models of typical fault behaviors and injects noise into normal historical data to synthesize large fault datasets [2]. This strategy is representative of much existing research: it focuses on per-sensor fault detection using synthetic or semi-synthetic data and leaves open how to exploit the rich spatio-temporal correlations that exist in real deployments, or how to translate fault information into more trustworthy multi-sensor fusion.

In parallel, the multi-sensor fusion and graph neural network (GNN) communities have developed tools that are highly relevant but not yet fully brought into Ag-IoT sensing. Credibility assessment methods such as Feng et al.’s multi-source heterogeneous fusion framework compute data-level and source-level credibility scores using statistical prediction models and neural networks, but are evaluated mainly on indoor and simulated datasets and use relatively small, manually defined graphs [3]. Evidence-theoretic approaches like Jiang et al.’s reliability-based basic belief assignment (BBA) explicitly incorporate source reliability into Dempster–Shafer fusion, yet rely on hand-crafted reliability measures and do not model high-dimensional spatio-temporal structure [4]. At the same time, spatio-temporal GNNs (ST-GNNs) have shown strong performance on graph-structured time series in traffic forecasting [5], industrial sensor anomaly detection [6], and frost prediction using an agricultural IoT platform plus nearby weather stations [7], but these models typically assume that input data are reliable and treat forecasting or anomaly detection as the end goal.

To the best of our knowledge, there is still no unified framework that (i) models an *in-field* Ag-IoT deployment as a dynamic spatio-temporal graph, (ii) learns node-level sensor credibility directly from multivariate time series, and (iii) uses these credibility estimates to drive reliability-aware fusion and imputation for downstream agricultural control. In this work, we move towards filling this gap by proposing a GNN-based, trust-aware spatio-temporal fusion

framework for Ag-IoT sensor networks. Our approach treats each sensor as a node in a spatio-temporal graph, learns latent dependencies among sensors across time, and outputs both fused environmental estimates and interpretable credibility scores that can be consumed by irrigation and environmental control logic.

2 Related Work

2.1 Sensor Fault Diagnosis in Ag-IoT

Sensor fault diagnosis in agricultural IoT has been systematically reviewed by Zou et al., who categorize existing methods into model-based, knowledge-based, and data-driven approaches, and emphasize the prevalence of sensor faults in harsh agricultural environments [1]. Data-driven methods based on machine learning and deep learning have recently received the most attention, as they can handle nonlinearity and complex multi-sensor interactions [1].

Within this line, Liang et al. propose an IDBO-SVM model tailored to agricultural environmental sensors, where an improved dung beetle optimization algorithm tunes SVM hyperparameters to diagnose bias, drift, accuracy degradation, and complete failure faults [2]. Their case study on Sichuan pepper fields shows that the method can reach an average diagnostic accuracy above 94% [2]. However, the work relies on simulating faults by adding noise to real historical data according to predefined mathematical models and ratios [2], and it focuses on classifying the state of individual sensors rather than on learning network-level trust patterns or integrating fault information into multi-sensor fusion.

2.2 Credibility-Aware Multi-Sensor Fusion

A second strand of related work quantifies the credibility or reliability of sensor data and integrates these estimates into fusion algorithms. Feng et al. present a credibility assessment method for multi-source heterogeneous sensor data that combines time-series prediction (ARIMA) and a backpropagation neural network to estimate both data-level and source-level credibility from spatio-temporal relationships among sensors [3]. Their experiments on a simulated smoke-temperature dataset and the Intel indoor wireless sensor network dataset demonstrate that explicit credibility modeling can improve the detection of unreliable measurements [3].

Complementary evidence-theoretic approaches incorporate source reliability into Dempster-Shafer fusion. For example, Jiang et al. propose a reliability-based BBA method that represents both uncertainty and source reliability and shows how this improves multi-sensor fusion decisions in classification tasks [4]. These techniques, however, generally operate on relatively small, static sensor sets and depend on hand-crafted reliability indicators or simple statistical predictors; they do not exploit the full spatio-temporal structure of large Ag-IoT

deployments or learn credibility in an end-to-end manner from raw multivariate time series.

2.3 Spatio-Temporal GNNs and GNNs in Smart Agriculture

Spatio-temporal GNNs have become a standard tool for modeling graph-structured time series. In traffic forecasting, Roy et al.’s Unified Spatio-Temporal Graph Convolutional Network (USTGCN) directly performs message passing on a space-time graph and outperforms factorized spatial-only and temporal-only baselines on multiple PeMS datasets [5]. In industrial settings, the MSDG model constructs a multi-scale dynamic graph over multivariate sensor time series and uses reconstruction error from a graph-based encoder-decoder to detect anomalies, outperforming conventional deep-learning baselines on several public datasets [6]. In agriculture-related climate applications, Lira et al. propose GRAST-Frost, a GNN with spatio-temporal attention that models an experimental IoT platform together with ten nearby weather stations to forecast minimum temperature and frost events, achieving superior performance over classical time-series and nongraph deep-learning models [7].

GNNs have also been used in smart agriculture for perception tasks. For instance, Lin et al. introduce GPA-Net, a graph-pyramid attention network that combines CNN backbones with graph-based attention and pyramid modules for fine-grained pest and disease identification from crop images, achieving strong accuracy on several benchmark datasets [8]. These works demonstrate that GNNs and ST-GNNs are well suited to capturing spatial relations and temporal dynamics in agricultural contexts. However, they typically assume reliable input data and target forecasting or classification performance as the primary objective, without explicitly modeling sensor credibility or integrating trust estimates into the fusion of dense in-field Ag-IoT sensor networks.

Our work is closest in spirit to credibility-aware sensor fusion and spatio-temporal GNNs, but it differs in that it treats an Ag-IoT deployment as a dynamic spatio-temporal graph and learns sensor credibility scores jointly with fusion, aiming specifically at improving the robustness of agricultural decision-making to sensor faults and degradation.

3 Technical Approach

3.1 Overview

We address the problem of reliable sensing in agricultural IoT deployments where low-cost sensors frequently drift, get stuck, or become noisy. Our core idea is to treat *sensor credibility* as a learnable quantity in a spatio-temporal graph: sensors are nodes, physical and statistical relations define edges, and a spatio-temporal graph neural network (ST-GNN) jointly produces a fused regional signal, per-sensor credibility scores, and anomaly indicators. During

fusion, low-credibility sensors are automatically down-weighted so that the final estimate remains trustworthy even when individual sensors fail.

3.2 Main Design

We divide the observation horizon into sequential sliding windows and, for each window, run our model on the corresponding multivariate time series segment.

Inputs.

- A sensor data tensor $\mathbf{X} \in R^{N \times W \times F}$, where N is the number of sensors, W is the number of time steps in the window, and F is the feature dimension (e.g., temperature, humidity, light).
- Sensor metadata: sensor coordinates (for spatial relations) and historical statistics (for correlation relations), used to construct the spatio-temporal graph.

Outputs.

- **Region-level fused time series** $\hat{\mathbf{Y}} \in R^{W \times F}$: a credible estimate of the environmental parameters for this region.
- **Sensor credibility** $\mathbf{C} \in [0, 1]^{N \times W}$: a time-varying credibility score for each sensor.
- **Anomaly indicators** $\mathbf{A} \in [0, 1]^{N \times W}$: per-sensor, per-time anomaly scores (or binary flags after thresholding).

3.3 Spatio-Temporal Graph and Model Design

We formalize the deployment as a dynamic graph whose nodes are sensors and whose edges reflect spatial and statistical relations.

- **Graph construction.** A spatial graph is built from coordinates using k -nearest neighbors or a distance threshold. A correlation graph is built from Pearson correlations of historical time series, connecting highly correlated sensors. In addition, an attention-based learnable graph refines edge weights online, so the overall adjacency combines spatial distance, correlation, and learned dependencies.
- **ST-GNN backbone.** For each sensor, a temporal encoder (e.g., temporal convolution or GRU) first extracts short-term dynamics from its windowed time series. Stacked spatio-temporal graph convolution layers then propagate information over the constructed graph, producing node embeddings that capture both local history and neighborhood context.

- **Credibility-aware fusion.** A credibility head maps each node embedding to a scalar $c_{i,t} \in (0, 1)$. A fusion head aggregates sensor information into the regional estimate using these scores as weights, so unreliable sensors contribute little to $\hat{\mathbf{Y}}$.
- **Anomaly scoring.** An anomaly head derives $a_{i,t}$ from how difficult each sensor is to reconstruct or predict and how inconsistent it is with its neighbors. High anomaly scores are encouraged to coincide with low credibility scores.

3.4 Data, Fault Injection, and Objective

We evaluate the method on two complementary data sources: (1) augmented open-source IoT datasets where we inject synthetic faults, and (2) an agricultural IoT testbed dataset collected in a controlled smart-farm-like environment. Across both, we inject representative fault modes, including stuck-at faults (constant readings), drift faults (slowly biased readings), random Gaussian noise, and missing blocks, with controlled start times and magnitudes.

Training minimizes a composite loss that balances three aspects: (i) fusion error between $\hat{\mathbf{Y}}$ and a reference environmental signal; (ii) violation of basic spatio-temporal and physical consistency (e.g., temperature–humidity should typically be negatively correlated, temperature–light often positively correlated during daytime); and (iii) a penalty that increases when many sensors must be assigned low credibility. Informally, the model is learned to minimize *fusion error* + *spatio-temporal inconsistency* + $1/\text{credibility}$, so that it learns to both detect problematic sensors and perform reliable graph-based data fusion.

4 Implementation and Expected Outcomes

4.1 Implementation Plan

The project will be implemented along two tightly coupled tracks: an offline software pipeline for method development and benchmarking, and an online hardware testbed for field validation.

On the software side, we will implement the credibility-aware ST-GNN framework in a standard deep learning stack (e.g., PyTorch). The pipeline will include: (i) data preprocessing and sliding-window construction of input tensors $\mathbf{X} \in \mathbb{R}^{N \times W \times F}$; (ii) automatic graph construction from sensor coordinates and historical correlations; (iii) model training and hyperparameter tuning on augmented open-source IoT datasets; and (iv) evaluation scripts for computing fusion error, anomaly detection metrics, and robustness under different fault scenarios.

On the hardware side, we will deploy a small-scale agricultural IoT testbed. ESP32-based sensing nodes will measure temperature, humidity, soil moisture, and light intensity, and transmit data to a Raspberry Pi 4 gateway. The gateway will run an optimized version of our ST-GNN (e.g., via model quantization or

ONNX export) to perform near real-time inference: computing region-level fused environmental time series, sensor credibility scores, and anomaly flags. This hardware–software co-design ensures that the proposed framework is not only effective in simulation, but also implementable on low-cost, resource-constrained platforms typical of agricultural deployments.

Validation will follow a dual-source methodology. First, we will augment open-source multivariate IoT datasets with representative fault modes, including stuck-at faults, drift injection, Gaussian noise, and missing blocks, to systematically stress-test the model under controlled conditions. Second, we will reproduce a subset of these fault patterns in the physical testbed (e.g., by inducing sensor freezing, miscalibration, or environmental perturbations) and compare online inference results with offline ground truth. Together, these two tracks bridge the gap between simulated robustness and practical deployability.

4.2 New Knowledge

The project is expected to generate several forms of new knowledge:

- **Quantitative understanding of credibility-aware fusion.** We will characterize how modeling per-sensor credibility in a spatio-temporal graph affects fusion error, anomaly detection accuracy, and robustness compared to naive averaging, static weighting, and non-graph baselines (e.g., RNN-based fusion).
- **Empirical analysis of sensor fault behaviours.** By injecting and detecting multiple fault types across datasets, we will obtain a systematic view of which fault modes are most harmful to fusion, how they manifest in spatio-temporal patterns, and which graph-based signals (self-history vs. neighborhood consistency vs. prediction deviation) are most informative.
- **Spatio-temporal consistency as a design prior.** We will evaluate simple physical priors (e.g., typical negative correlation between temperature and humidity, positive daytime correlation between temperature and light) as regularizers in the loss. This will reveal to what extent such coarse domain knowledge can improve robustness in noisy, real-world agricultural settings.
- **Graph design guidelines for sparse sensor networks.** Through ablation on spatial, correlation-based, and learnable edges, we expect to derive practical guidelines on graph topology (e.g., choice of k -nearest neighbors, correlation thresholds, and dynamic attention) for low-density agricultural IoT deployments.

4.3 Practical Applications and Deliverables

Beyond methodological insights, the project will produce concrete artefacts that can be reused or extended:

- **An open-source implementation of credibility-aware ST-GNN fusion.** This includes modular code for graph construction, model training, and evaluation, which can be adapted to other IoT domains (e.g., smart buildings or environmental monitoring).
- **Annotated benchmark datasets with fault injections.** We will release processed versions of the augmented IoT datasets and, where feasible, anonymised testbed logs, with labelled fault intervals and ground-truth fusion targets to facilitate future comparisons.
- **A reference edge deployment blueprint.** The ESP32 + Raspberry Pi 4 stack, along with configuration details and resource measurements (latency, memory, power), will serve as a template for low-cost agricultural IoT deployments requiring trustworthy sensing.
- **Prototype monitoring interface.** A lightweight dashboard (e.g., web-based) will visualise fused environmental time series, per-sensor credibility, and anomaly alerts, demonstrating how farmers or system operators could interact with the model in practice.

Taken together, these outcomes move beyond passive data collection toward *trust-aware* agricultural sensing: instead of fusing all sensor readings indiscriminately, the system continuously certifies which sensors are reliable, how much they should influence the fused estimate, and when human attention is needed due to suspected faults or inconsistent measurements.

5 Description of Each Member’s Responsibilities

The project will be carried out collaboratively by the two members with responsibilities divided as follows:

- **Bohan Liu (Project Lead):** Overall project design and coordination. Primary responsibility for the GNN-based fusion algorithm, credibility modeling, and edge deployment framework.
- **Yihang Su:** Data pipeline construction, fault injection strategies, spatio-temporal graph implementation, and comprehensive system evaluation and benchmarking.

Both members will contribute jointly to software development, testbed deployment, and the preparation of the final report and deliverables.

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